Mobile Application Development Lab Record [ICT 3268]

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Roll No: 40

CCE-C2

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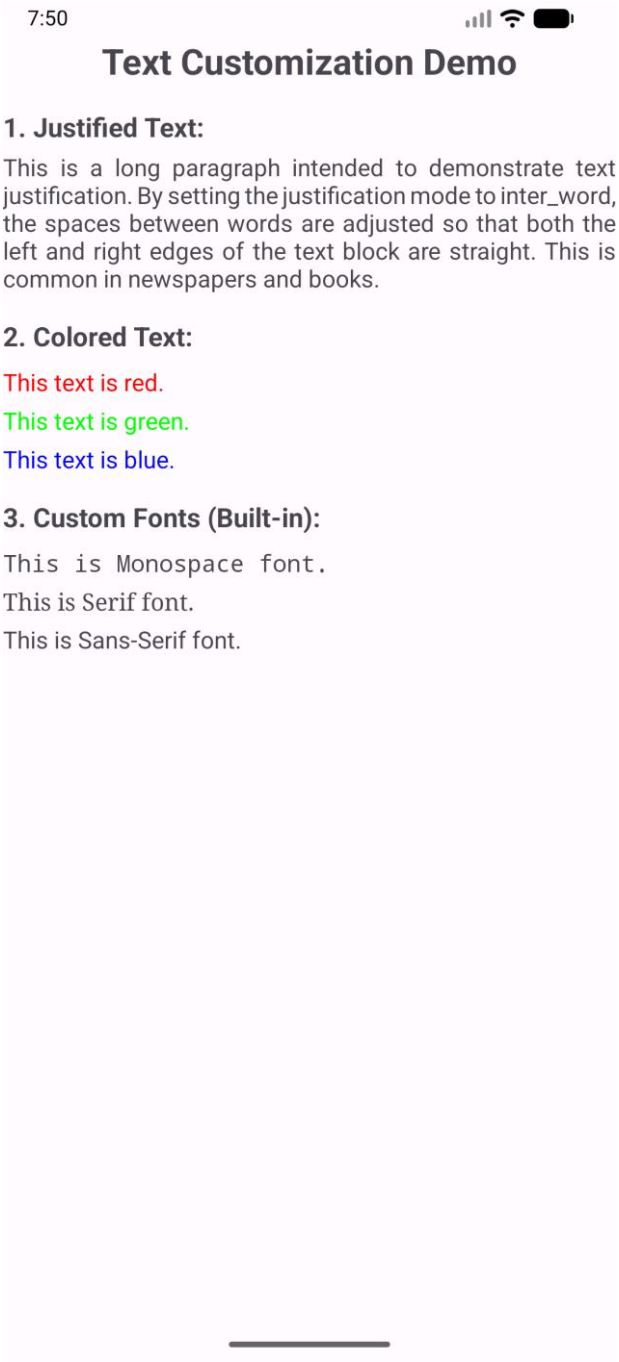
LAB NO: 1

BASICS OF ANDROID MOBILE APPLICATION DEVELOPMENT TOOL

Develop an Android application for the following programs:

1. Create an Android application to show the demo of displaying text with justifying elements, changing text colors ,fonts etc.



activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="16dp"
  tools:context=".MainActivity">
  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Text Styling Demo"
    android:textSize="24sp"
    android:textStyle="bold"
    android:layout_marginBottom="16dp"
    android:gravity="center"/>
  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Justified Text Example:"
    android:textStyle="bold"
    android:layout_marginBottom="4dp"
    android:textSize="18sp"/>

  <TextView
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="This is a long paragraph intended to demonstrate the justification mode inter_word. It will stretch the spaces between words to ensure both left and right edges are aligned."
    android:justificationMode="inter_word"
    android:layout_marginBottom="16dp"
    android:textSize="16sp"/>
  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Colored Text:"
    android:textStyle="bold"
    android:layout_marginBottom="8dp"
    android:textSize="18sp"/>
  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="This text is Red"
    android:textColor="#FF0000"
    android:layout_marginBottom="4dp"
    android:textSize="16sp"/>
  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="This text is Green"
    android:textColor="#00FF00"
    android:layout_marginBottom="4dp"
    android:textSize="16sp"/>
```

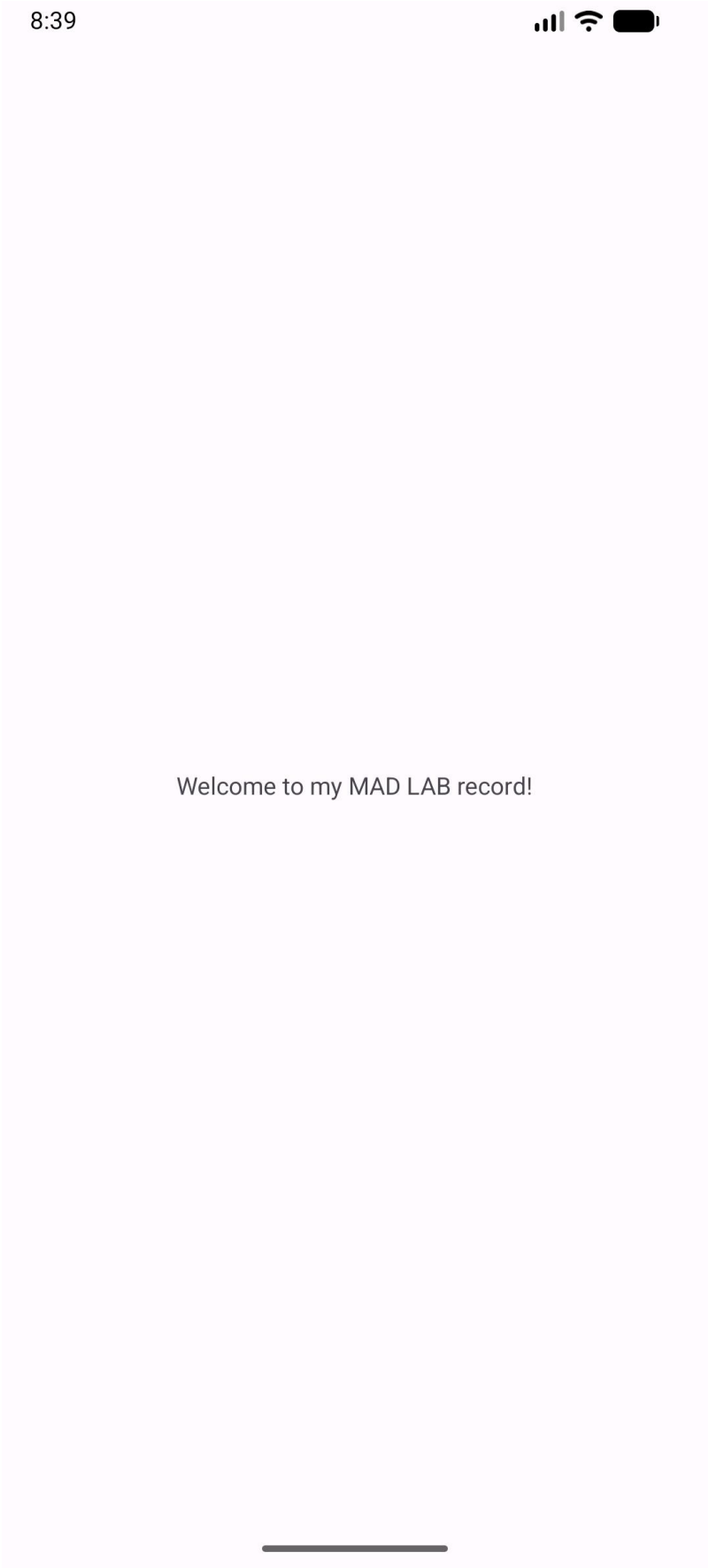
```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="This text is Blue"
    android:textColor="#0000FF"
    android:layout_marginBottom="16dp"
    android:textSize="16sp"/>
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Font Families:"
    android:textStyle="bold"
    android:textSize="18sp"
    android:layout_marginBottom="8dp"/>
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="This is Monospace Font"
    android:fontFamily="monospace"
    android:layout_marginBottom="4dp"
    android:textSize="16sp"/>
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="This is Serif Font"
    android:fontFamily="serif"
    android:layout_marginBottom="4dp"
    android:textSize="16sp"/>
```

```
    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="This is Sans-Serif Font"
        android:fontFamily="sans-serif"
        android:layout_marginBottom="4dp"
        android:textSize="16sp"/>
</LinearLayout>
```

MainActivity.java:

```
package com.example.textdemo;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

2. Find the “hello word” text in the XML document and modify the text.



activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity">

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="@string/hello_android"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java:

```
package com.example.myapplication2;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    EdgeToEdge.enable(this);
    setContentView(R.layout.activity_main);
```

```
    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets)
-> {
        Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
        v.setPadding(systemBars.left, systemBars.top, systemBars.right,
systemBars.bottom);
        return insets;
    });
  }
}
```

LAB NO: 2

INTRODUCTION TO ACTIVITY AND LAYOUTS IN ANDROID

1. Create an app that illustrates that the activity lifecycle method being triggered by various action. Understand when onCreate(), onStart(), onResume event occur.

8:55
Activity Lifecycle Demo
Check Logcat or Toasts to see lifecycle transitions.
onResume called

activity_main.xml:

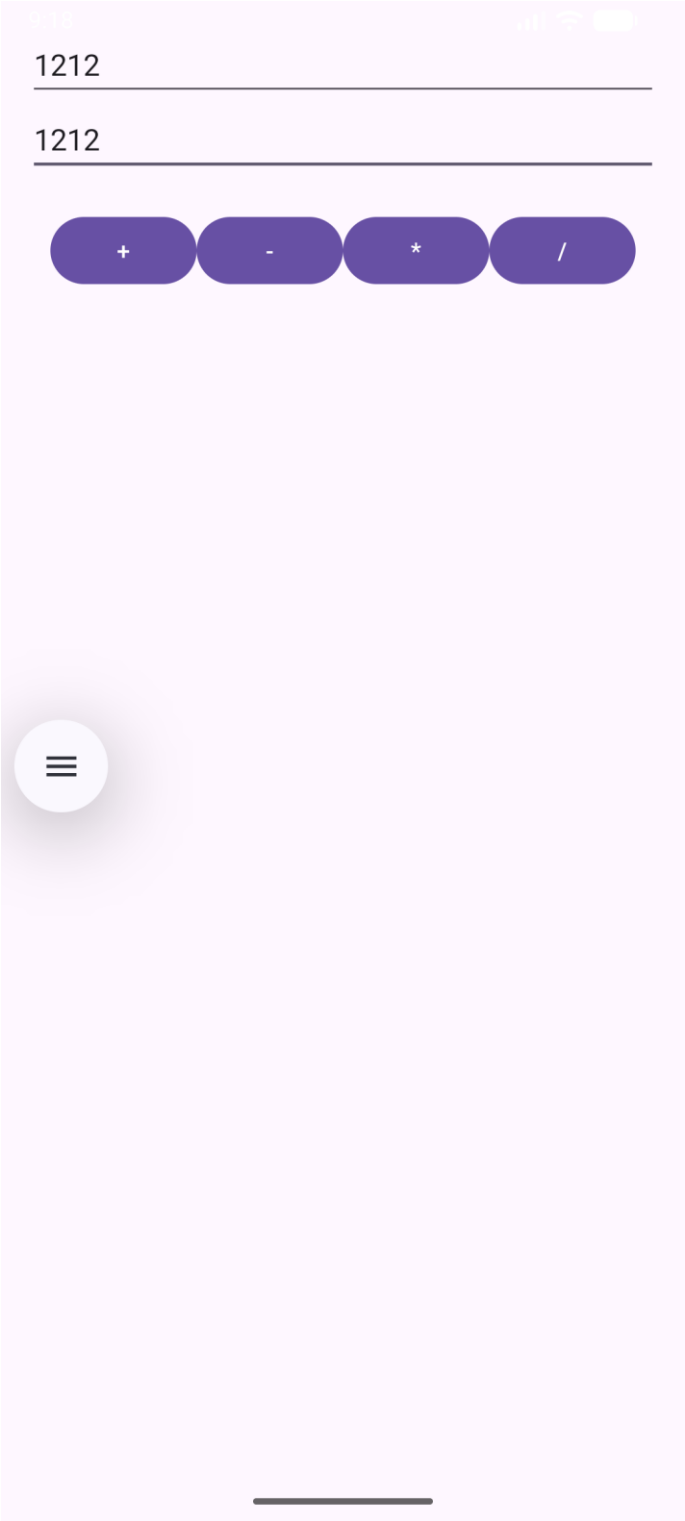
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="16dp">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Activity Lifecycle Demo"
      android:textSize="24sp"
      android:textStyle="bold" />

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Check Logcat or Toasts to see lifecycle transitions."
      android:layout_marginTop="10dp"
      android:textAlignment="center"/>

</LinearLayout>
```

MainActivity.java:

```
package com.example.myapplication1;

import android.os.Bundle;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

   @Override
   protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
```

```
        showToast("onCreate called"); // Initializing the activity
    }

    @Override
    protected void onStart() {
        super.onStart();
        showToast("onStart called"); // Activity is becoming visible
    }

    @Override
    protected void onResume() {
        super.onResume();
        showToast("onResume called"); // Activity is now in the foreground
    }

    @Override
    protected void onPause() {
        super.onPause();
        showToast("onPause called"); // Activity is partially obscured
    }

    @Override
    protected void onStop() {
        super.onStop();
        showToast("onStop called"); // Activity is no longer visible
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        showToast("onDestroy called"); // Final cleanup [cite: 295]
    }

    // Helper method to display Toast messages [cite: 640]
    private void showToast(String message) {
        Toast.makeText(this, message, Toast.LENGTH_SHORT).show();
    }
}
```

2. Create a Calculator app that does the function of multiplication, addition, division, subtraction but displays the result in the format of:-Num1 operator num2 = result. Back button on the next activity should get back to the calculator activity again.



activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="16dp">

  <EditText
    android:id="@+id/num1"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Number 1"
    android:inputType="numberDecimal" />

  <EditText
    android:id="@+id/num2"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Number 2"
    android:inputType="numberDecimal" />

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center"
    android:layout_marginTop="20dp">

    <Button android:id="@+id/btnAdd" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="+" />
    <Button android:id="@+id/btnSub" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="-" />
    <Button android:id="@+id/btnMul" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="*" />
    <Button android:id="@+id/btnDiv" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="/" />
  </LinearLayout>

</LinearLayout>
```

activity_result.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:gravity="center"
  android:orientation="vertical">

  <TextView
    android:id="@+id/resultText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textSize="24sp"
    android:textStyle="bold" />

</LinearLayout>
```

ANDROID MANIFEST XML CODE (add this to manifest):

```
<activity
  android:name=".ResultActivity"
  android:exported="false" />
```

MainActivity.java:

```
package com.example.myapplication;

import android.content.Intent;
import android.os.Bundle;
import android.widget.EditText;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {
  EditText n1, n2;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
```

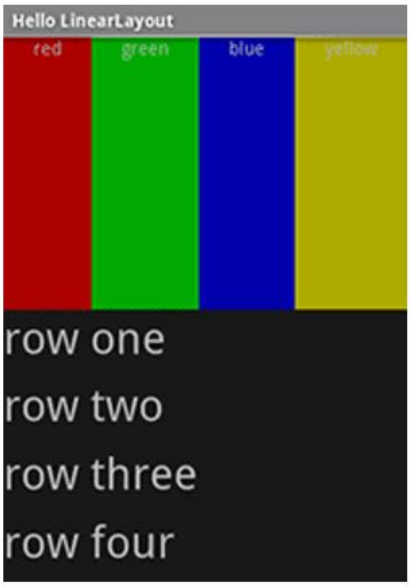
```
        setContentView(R.layout.activity_main);

        n1 = findViewById(R.id.num1);
        n2 = findViewById(R.id.num2);

        findViewById(R.id.btnAdd).setOnClickListener(v -> calculate('+'));
        findViewById(R.id.btnSub).setOnClickListener(v -> calculate('-'));
        findViewById(R.id.btnMul).setOnClickListener(v -> calculate('*'));
        findViewById(R.id.btnDiv).setOnClickListener(v -> calculate('/'));
    }

    private void calculate(char op) {
        double v1 = Double.parseDouble(n1.getText().toString());
        double v2 = Double.parseDouble(n2.getText().toString());
        double res = 0;

        if (op == '+') res = v1 + v2;
        else if (op == '-') res = v1 - v2;
        else if (op == '*') res = v1 * v2;
        else if (op == '/') res = v1 / v2;

        String out = v1 + " " + op + " " + v2 + " = " + res;

        Intent intent = new Intent(this, ResultActivity.class);
        intent.putExtra("RESULT_KEY", out); // Passing data using key-value pairs
[cite: 727]
        startActivity(intent);
    }
}
```

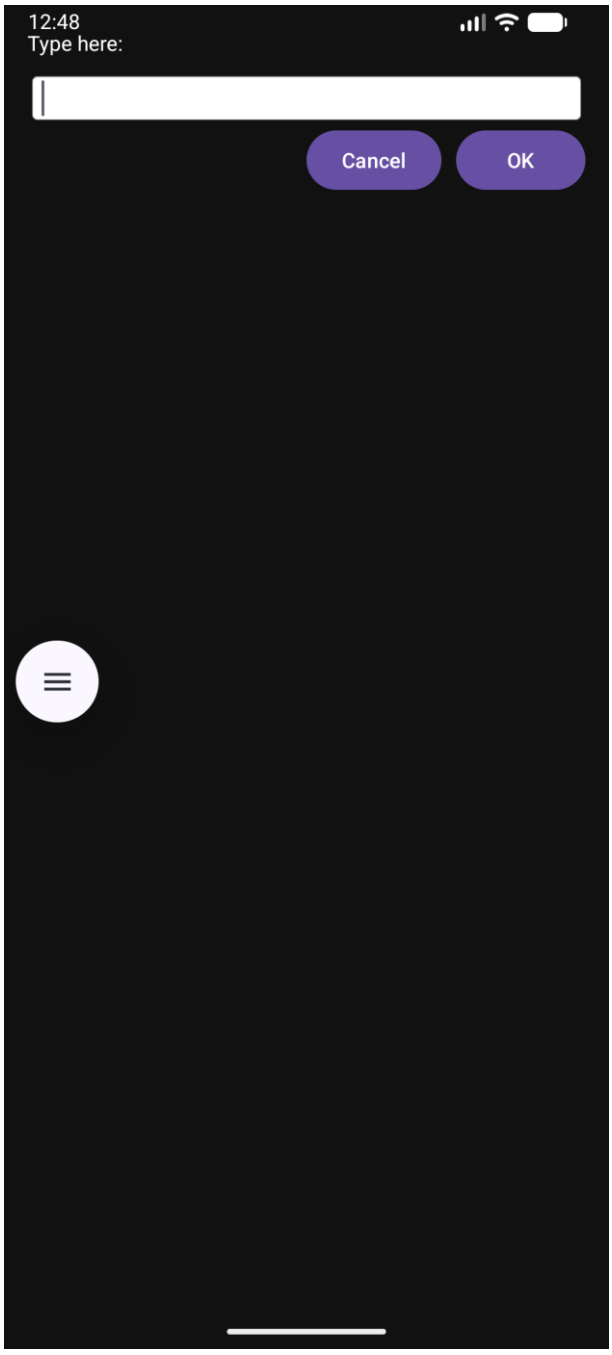
ResultActivity.java:

```
package com.example.myapplication;
import android.os.Bundle;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;
public class ResultActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_result);

        String result = getIntent().getStringExtra("RESULT_KEY");
        TextView tv = findViewById(R.id.resultText);
        tv.setText(result);
    }
}
```

3. Create the following given scenario using linear and relative layout concept.





layout_linear_scenario.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical">

  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:background="#808080"
    android:padding="5dp"
    android:text="Hello LinearLayout"
    android:textColor="#FFFFFF" />

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    android:orientation="horizontal">

    <TextView
      android:layout_width="0dp"
      android:layout_height="match_parent"
      android:layout_weight="1"
      android:background="#FF0000"
      android:gravity="center_horizontal"
      android:text="red" />

    <TextView
      android:layout_width="0dp"
      android:layout_height="match_parent"
      android:layout_weight="1"
      android:background="#00FF00"
      android:gravity="center_horizontal"
      android:text="green" />
```

```
    <TextView
      android:layout_width="0dp"
      android:layout_height="match_parent"
      android:layout_weight="1"
      android:background="#0000FF"
      android:gravity="center_horizontal"
      android:text="blue" />

    <TextView
      android:layout_width="0dp"
      android:layout_height="match_parent"
      android:layout_weight="1"
      android:background="#FFFF00"
      android:gravity="center_horizontal"
      android:text="yellow" />
  </LinearLayout>

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1"
    android:background="#222222"
    android:orientation="vertical"
    android:padding="10dp">

    <TextView android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="row one"
android:textColor="#FFFFFF" android:textSize="24sp" />
    <TextView android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="row two"
android:textColor="#FFFFFF" android:textSize="24sp" />
    <TextView android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="row three"
android:textColor="#FFFFFF" android:textSize="24sp" />
    <TextView android:layout_width="wrap_content"
android:layout_height="wrap_content" android:text="row four"
android:textColor="#FFFFFF" android:textSize="24sp" />
  </LinearLayout>
</LinearLayout>
```

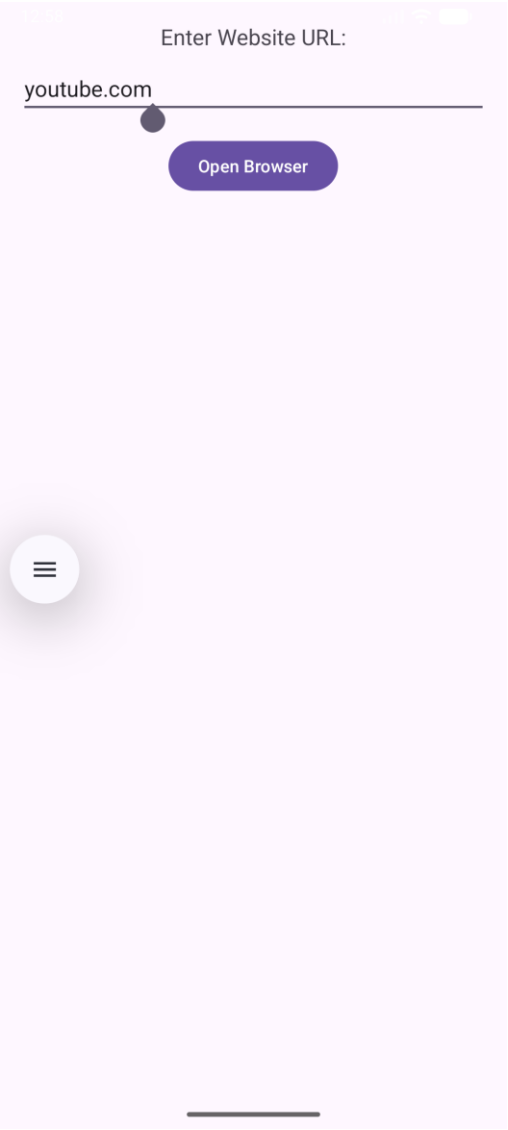
layout_relative_scenario.xml:

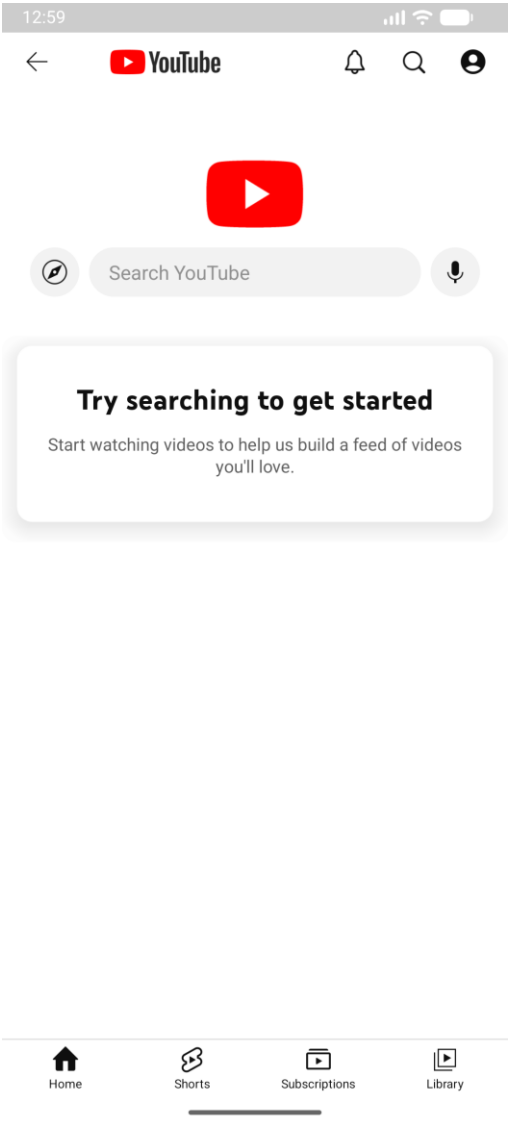
```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:background="#111111"
  android:padding="16dp">

  <TextView
    android:id="@+id/title"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Type here:"
    android:textColor="#FFFFFF" />

  <EditText
    android:id="@+id/input"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@id/title"
    android:background="@android:drawable/editbox_background_normal"
    android:layout_marginTop="10dp" />

  <Button
    android:id="@+id/btnOk"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/input"
    android:layout_alignParentEnd="true"
    android:text="OK" />

  <Button
    android:id="@+id/btnCancel"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_below="@id/input"
    android:layout_toStartOf="@id/btnOk"
    android:layout_marginEnd="10dp"
    android:text="Cancel" />
</RelativeLayout>
```

MainAcitivity.java:

```
package com.example.myapplication;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);

    // Change this line to switch between relative and linear layout scenarios for testing
    setContentView(R.layout.layout_relative_scenario);
  }
}
```

4. Create an app such that when the user click on the given URL typed by the user, it visits the corresponding page.





activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="16dp"
   android:gravity="center_horizontal">

   <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Enter Website URL:"
      android:textSize="18sp"
      android:layout_marginBottom="8dp"/>

   <EditText
      android:id="@+id/urlInput"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="https://www.google.com"
      android:inputType="textUri"
      android:layout_marginBottom="16dp" />

   <Button
      android:id="@+id/btnVisit"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Open Browser" />

</LinearLayout>
```

MainActivity.java:

```
package com.example.myapplication;

import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.widget.Button;
```

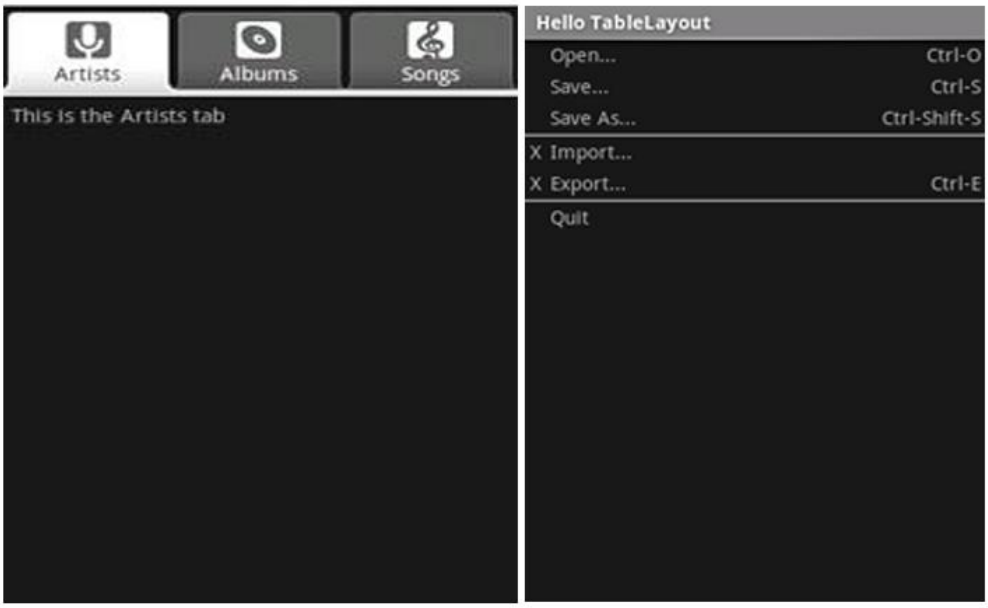
```java
import android.widget.EditText;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Link the XML components to Java
        EditText urlInput = findViewById(R.id.urlInput);
        Button btnVisit = findViewById(R.id.btnVisit);

        btnVisit.setOnClickListener(view -> {
            String url = urlInput.getText().toString().trim();

            if (!url.isEmpty()) {
                // Ensure the URL has a protocol, otherwise Uri.parse may fail
                if (!url.startsWith("http://") && !url.startsWith("https://")) {
                    url = "https://" + url;
                }

                try {
                    // Create an Implicit Intent to view the URL
                    Intent intent = new Intent(Intent.ACTION_VIEW, Uri.parse(url));
                    startActivity(intent);
                } catch (Exception e) {
                    // Handle cases where no browser is installed or URL is invalid
                    Toast.makeText(this, "Could not open link", Toast.LENGTH_SHORT).show();
                }
            } else {
                Toast.makeText(this, "Please enter a URL", Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

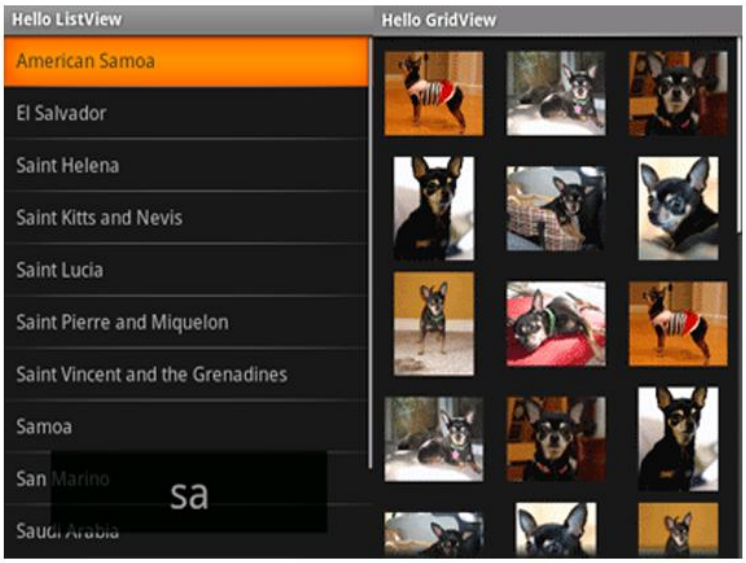
LAB NO: 3

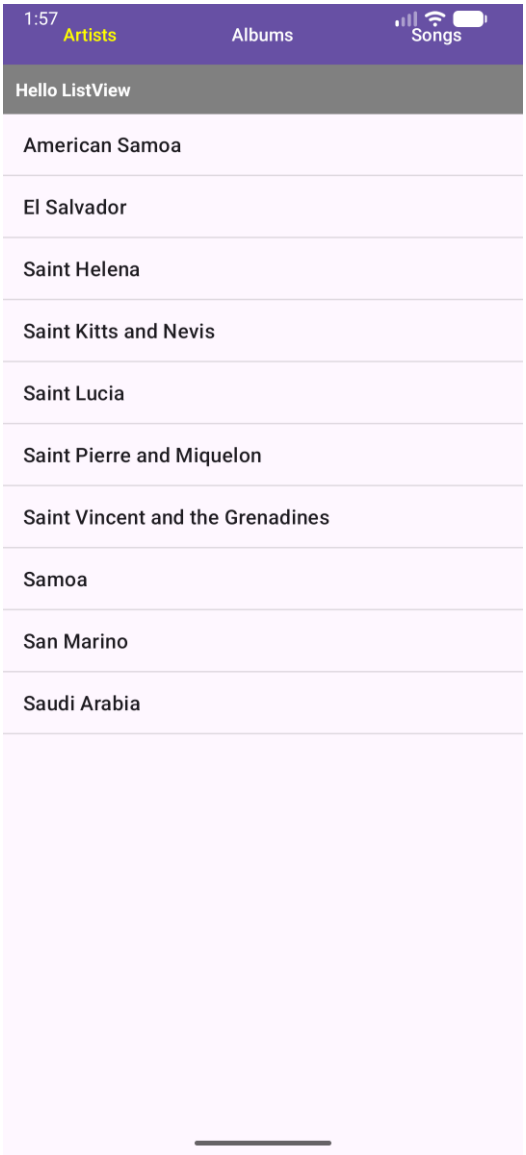
ACTIVITY AND LAYOUT CONTINUED

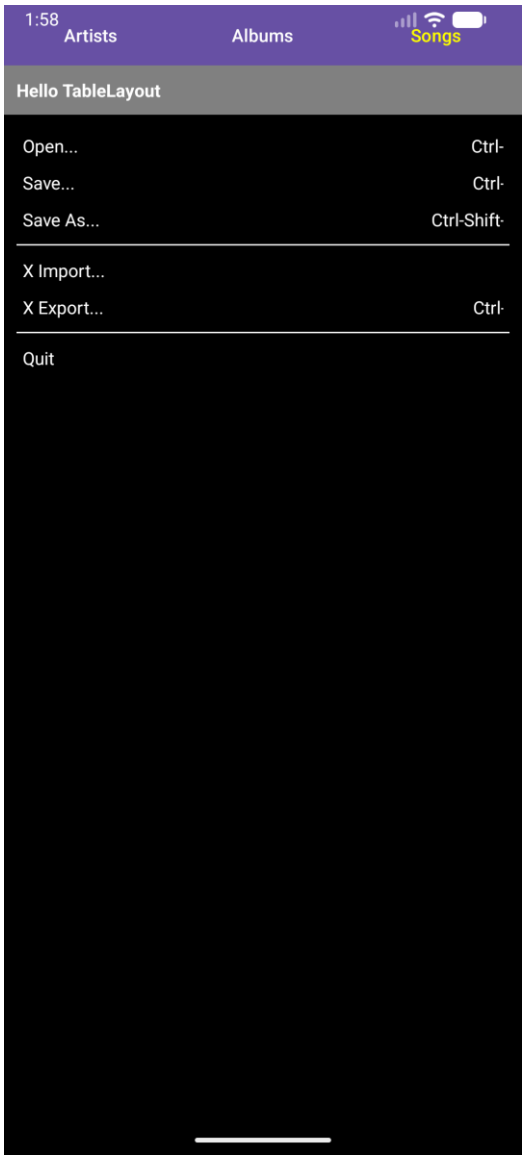
Lab Exercise:

1. Using given scenario develop an application to perform following layout operations
   - List view
   - Grid view
   - Tab layout
   - Table layout









activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical">

  <com.google.android.material.tabs.TabLayout
    android:id="@+id/tabLayout"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
```

```
        android:background="?attr/colorPrimary"
        app:tabTextColor="#FFFFFF"
        app:tabSelectedTextColor="#FFFF00"
        app:tabMode="fixed"
        app:tabGravity="fill" />

    <androidx.viewpager.widget.ViewPager
        android:id="@+id/viewPager"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

fragment_list.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Hello ListView"
        android:background="#808080"
        android:padding="10dp"
        android:textColor="#FFFFFF"
        android:textStyle="bold"/>

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />
</LinearLayout>
```

fragment_grid.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Hello GridView"
        android:background="#808080"
        android:padding="10dp"
        android:textColor="#FFFFFF"
        android:textStyle="bold"/>

    <GridView
        android:id="@+id/gridView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:numColumns="3"
        android:horizontalSpacing="10dp"
        android:verticalSpacing="10dp"
        android:padding="10dp"
        android:gravity="center"/>
</LinearLayout>
```

fragment_table.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="#000000">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Hello TableLayout"
        android:background="#808080"
        android:padding="10dp"
        android:textColor="#FFFFFF"
```

```
    android:textStyle="bold"/>

  <TableLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:stretchColumns="1"
    android:padding="10dp">

    <TableRow android:padding="5dp">
      <TextView android:text="Open..." android:textColor="#FFFFFF"/>
      <TextView android:text="Ctrl-O" android:gravity="end"
android:textColor="#FFFFFF"/>
    </TableRow>

    <TableRow android:padding="5dp">
      <TextView android:text="Save..." android:textColor="#FFFFFF"/>
      <TextView android:text="Ctrl-S" android:gravity="end"
android:textColor="#FFFFFF"/>
    </TableRow>

    <TableRow android:padding="5dp">
      <TextView android:text="Save As..." android:textColor="#FFFFFF"/>
      <TextView android:text="Ctrl-Shift-S" android:gravity="end"
android:textColor="#FFFFFF"/>
    </TableRow>

    <View android:layout_height="1dp" android:background="#FFFFFF"
android:layout_marginTop="5dp" android:layout_marginBottom="5dp"/>

    <TableRow android:padding="5dp">
      <TextView android:text="X Import..." android:textColor="#FFFFFF"/>
    </TableRow>

    <TableRow android:padding="5dp">
      <TextView android:text="X Export..." android:textColor="#FFFFFF"/>
      <TextView android:text="Ctrl-E" android:gravity="end"
android:textColor="#FFFFFF"/>
    </TableRow>

    <View android:layout_height="1dp" android:background="#FFFFFF"
android:layout_marginTop="5dp" android:layout_marginBottom="5dp"/>
```

```
    <TableRow android:padding="5dp">
      <TextView android:text="Quit" android:textColor="#FFFFFF"/>
    </TableRow>
  </TableLayout>
</LinearLayout>
```

MainActivity.java:

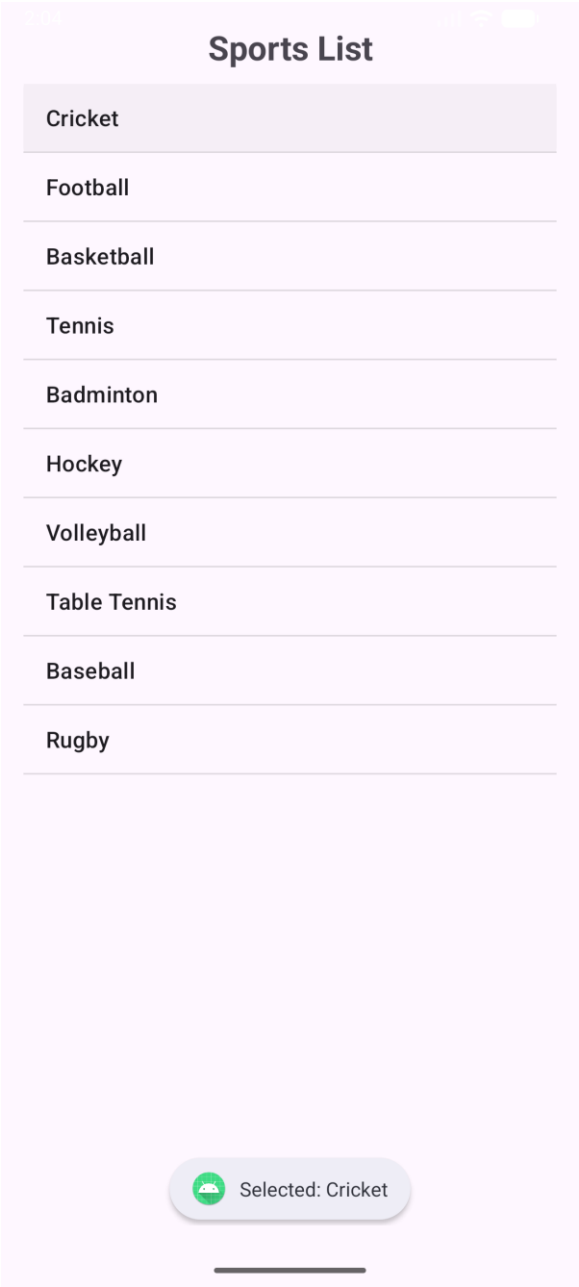
```
package com.example.myapplication;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ArrayAdapter;
import android.widget.GridView;
import android.widget.ListView;

import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentPagerAdapter;
import androidx.viewpager.widget.ViewPager;

import com.google.android.material.tabs.TabLayout;

import java.util.ArrayList;
import java.util.List;

public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
```

```
        TabLayout tabLayout = findViewById(R.id.tabLayout);
        ViewPager viewPager = findViewById(R.id.viewPager);

        // Set up the ViewPager with the sections adapter
        ViewPagerAdapter adapter = new
ViewPagerAdapter(getSupportFragmentManager());

        // Adding Fragments for each Tab
        adapter.addFragment(new ListFragment(), "Artists"); // List View Tab
        adapter.addFragment(new GridFragment(), "Albums");  // Grid View Tab
        adapter.addFragment(new TableFragment(), "Songs");  // Table Layout Tab

        viewPager.setAdapter(adapter);
        tabLayout.setupWithViewPager(viewPager);
    }

    // Adapter Class to Manage Fragments
    class ViewPagerAdapter extends FragmentPagerAdapter {
        private final List<Fragment> mFragmentList = new ArrayList<>();
        private final List<String> mFragmentTitleList = new ArrayList<>();

        public ViewPagerAdapter(FragmentManager manager) {
            super(manager, BEHAVIOR_RESUME_ONLY_CURRENT_FRAGMENT);
        }

        @Override
        public Fragment getItem(int position) {
            return mFragmentList.get(position);
        }

        @Override
        public int getCount() {
            return mFragmentList.size();
        }

        public void addFragment(Fragment fragment, String title) {
            mFragmentList.add(fragment);
            mFragmentTitleList.add(title);
        }

        @Override
```

```
    public CharSequence getPageTitle(int position) {
        return mFragmentTitleList.get(position);
    }
}

// 1. LIST FRAGMENT
public static class ListFragment extends Fragment {
    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater, @Nullable ViewGroup
container, @Nullable Bundle savedInstanceState) {
        View view = inflater.inflate(R.layout.fragment_list, container, false);
        ListView listView = view.findViewById(R.id.listView);

        // Data for ListView [cite: 574]
        String[] countries = {
            "American Samoa", "El Salvador", "Saint Helena",
            "Saint Kitts and Nevis", "Saint Lucia",
            "Saint Pierre and Miquelon", "Saint Vincent and the Grenadines",
            "Samoa", "San Marino", "Saudi Arabia"
        };

        ArrayAdapter<String> adapter = new ArrayAdapter<>(getContext(),
android.R.layout.simple_list_item_1, countries);
        listView.setAdapter(adapter);
        return view;
    }
}

// 2. GRID FRAGMENT
public static class GridFragment extends Fragment {
    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater, @Nullable ViewGroup
container, @Nullable Bundle savedInstanceState) {
        View view = inflater.inflate(R.layout.fragment_grid, container, false);
        GridView gridView = view.findViewById(R.id.gridView);

        // Using dummy text data for grid to avoid missing image errors
        String[] data = {"Img 1", "Img 2", "Img 3", "Img 4", "Img 5", "Img 6", "Img 7", "Img 8",
"Img 9"};
```

```
        ArrayAdapter<String> adapter = new ArrayAdapter<>(getContext(),
android.R.layout.simple_list_item_1, data);
        gridView.setAdapter(adapter);
        return view;
    }
  }

  // 3. TABLE FRAGMENT
  public static class TableFragment extends Fragment {
    @Nullable
    @Override
    public View onCreateView(@NonNull LayoutInflater inflater, @Nullable ViewGroup
container, @Nullable Bundle savedInstanceState) {
        // Just inflating the static table layout [cite: 563]
        return inflater.inflate(R.layout.fragment_table, container, false);
    }
  }
}
```

2. Write a program to list different sports, and when a sport is selected, display a message showing the selected sport.



Activity_main.xml:

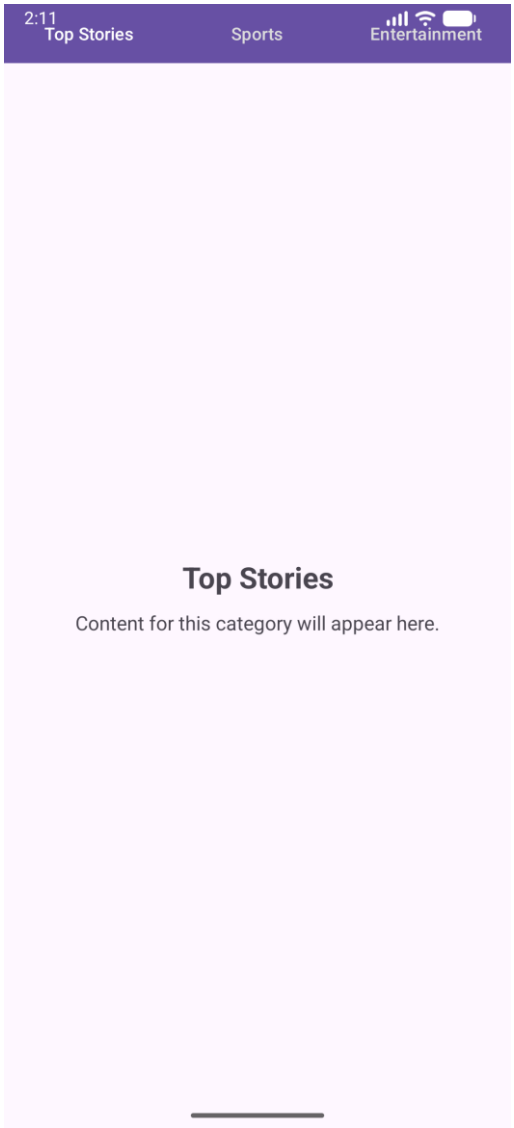
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="16dp">

  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Sports List"
    android:textSize="24sp"
    android:textStyle="bold"
    android:layout_marginBottom="10dp"
    android:gravity="center"/>

  <ListView
    android:id="@+id/sportsListView"
    android:layout_width="match_parent"
    android:layout_height="match_parent" />

</LinearLayout>
```

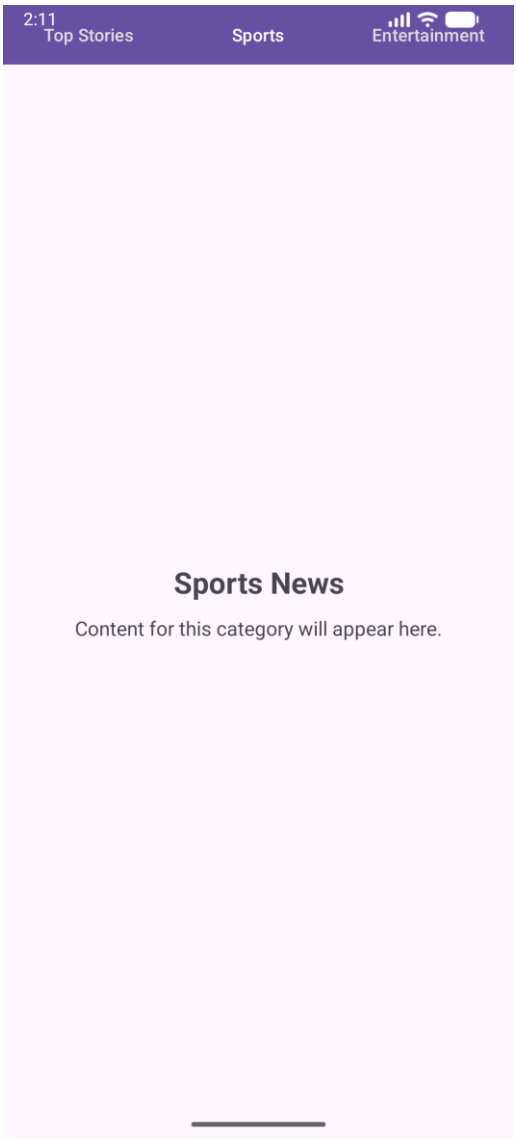
MainActivity.java:

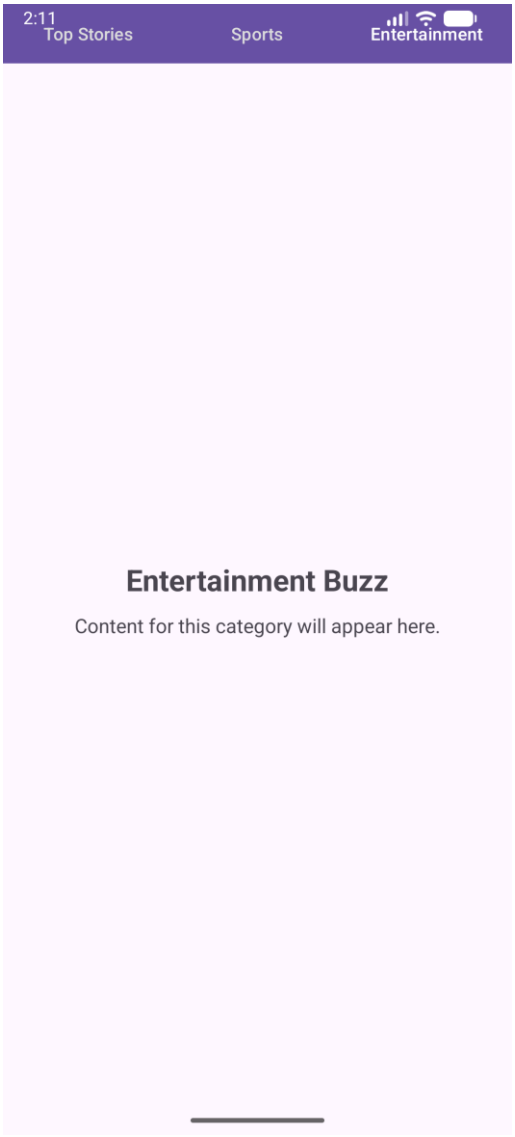
```
package com.example.myapplication;

import android.os.Bundle;
import android.view.View;
import android.widget.AdapterView;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
```

```
    // 1. Initialize the ListView
    ListView listView = findViewById(R.id.sportsListView);

    // 2. Define the data source (Array of Sports)
    String[] sports = {
        "Cricket", "Football", "Basketball", "Tennis",
        "Badminton", "Hockey", "Volleyball", "Table Tennis",
        "Baseball", "Rugby"
    };

    // 3. Create an ArrayAdapter to bind data to the ListView [cite: 415]
    ArrayAdapter<String> adapter = new ArrayAdapter<>(
        this,
        android.R.layout.simple_list_item_1,
        sports
    );

    // 4. Set the adapter to the ListView [cite: 416]
    listView.setAdapter(adapter);

    // 5. Set an OnItemClickListener to handle clicks
    listView.setOnItemClickListener(new AdapterView.OnItemClickListener() {
      @Override
      public void onItemClick(AdapterView<?> parent, View view, int position,
long id) {
        // Get the selected item text
        String selectedSport = sports[position];

        // Display the selected sport in a Toast message [cite: 635]
        Toast.makeText(MainActivity.this, "Selected: " + selectedSport,
Toast.LENGTH_SHORT).show();
      }
    });
  }
}
```

3. Design an news application and Implement the navigation between sections like Top Stories, Sports, and Entertainment using tab layout.







Activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   xmlns:app="http://schemas.android.com/apk/res-auto"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical">
```

```
    <com.google.android.material.tabs.TabLayout
        android:id="@+id/tabLayout"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:background="?attr/colorPrimary"
        app:tabTextColor="#DDDDDD"
        app:tabSelectedTextColor="#FFFFFF"
        app:tabMode="fixed"
        app:tabGravity="fill" />

    <androidx.viewpager.widget.ViewPager
        android:id="@+id/viewPager"
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

</LinearLayout>
```

Fragment_new.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/sectionLabel"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="News Section"
        android:textSize="24sp"
        android:textStyle="bold"
        android:gravity="center"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
    android:text="Content for this category will appear here."
    android:layout_marginTop="10dp"
    android:textSize="16sp"/>

</LinearLayout>
```

MainActivity.java:

```
package com.example.myapplication;

import android.os.Bundle;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.Fragment;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentPagerAdapter;
import androidx.viewpager.widget.ViewPager;
import com.google.android.material.tabs.TabLayout;
import java.util.ArrayList;
import java.util.List;

public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    TabLayout tabLayout = findViewById(R.id.tabLayout);
    ViewPager viewPager = findViewById(R.id.viewPager);

    // Set up the adapter
    ViewPagerAdapter adapter = new
ViewPagerAdapter(getSupportFragmentManager());

    // Add Fragments for each required section
    adapter.addFragment(new NewsFragment("Top Stories"), "Top Stories");
    adapter.addFragment(new NewsFragment("Sports News"), "Sports");
    adapter.addFragment(new NewsFragment("Entertainment Buzz"), "Entertainment");
```

```
        // Link ViewPager and TabLayout
        viewPager.setAdapter(adapter);
        tabLayout.setupWithViewPager(viewPager);
    }

    // Adapter Class to Manage Tabs
    static class ViewPagerAdapter extends FragmentPagerAdapter {
        private final List<Fragment> mFragmentList = new ArrayList<>();
        private final List<String> mFragmentTitleList = new ArrayList<>();

        public ViewPagerAdapter(FragmentManager manager) {
            super(manager, BEHAVIOR_RESUME_ONLY_CURRENT_FRAGMENT);
        }

        @NonNull
        @Override
        public Fragment getItem(int position) {
            return mFragmentList.get(position);
        }

        @Override
        public int getCount() {
            return mFragmentList.size();
        }

        public void addFragment(Fragment fragment, String title) {
            mFragmentList.add(fragment);
            mFragmentTitleList.add(title);
        }

        @Nullable
        @Override
        public CharSequence getPageTitle(int position) {
            return mFragmentTitleList.get(position);
        }
    }
}
```

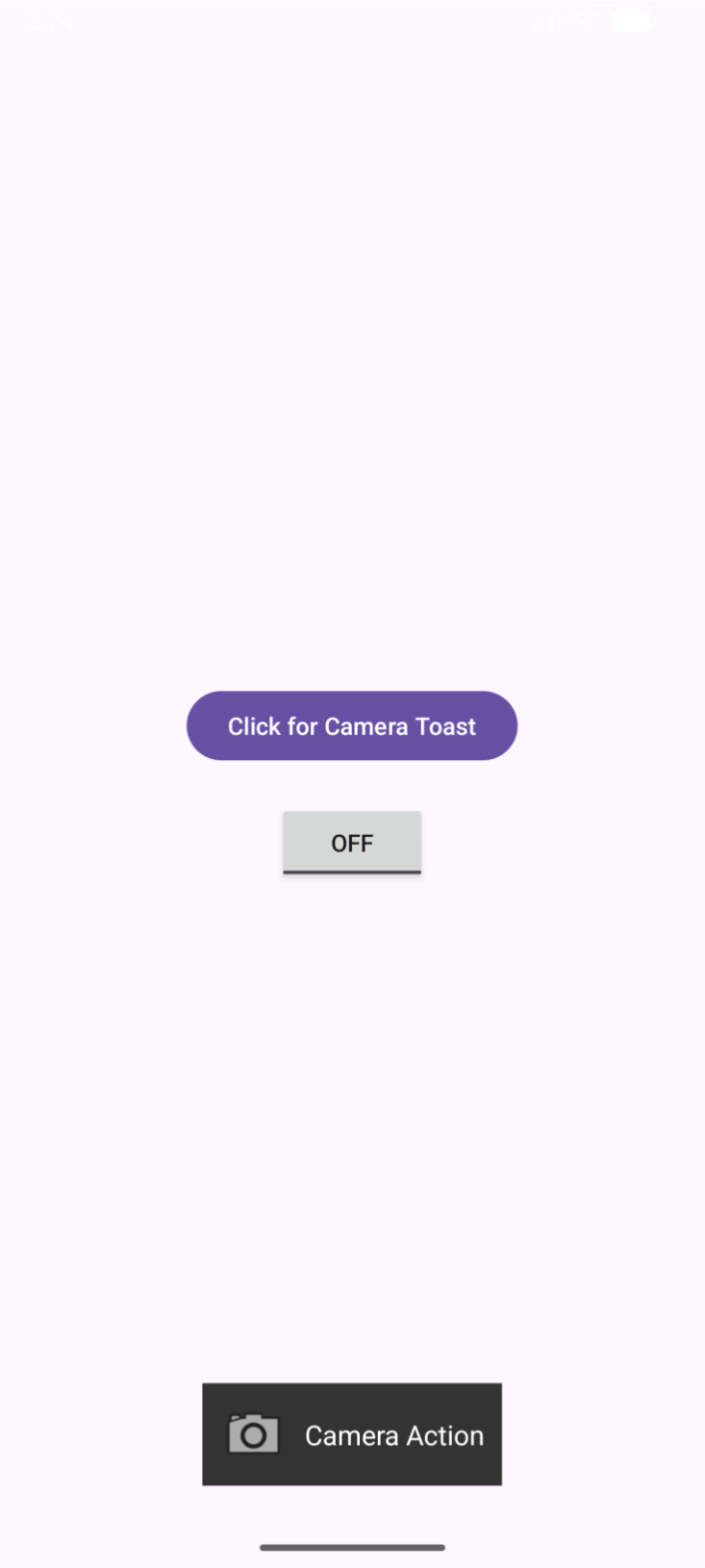
NewFragment.java:

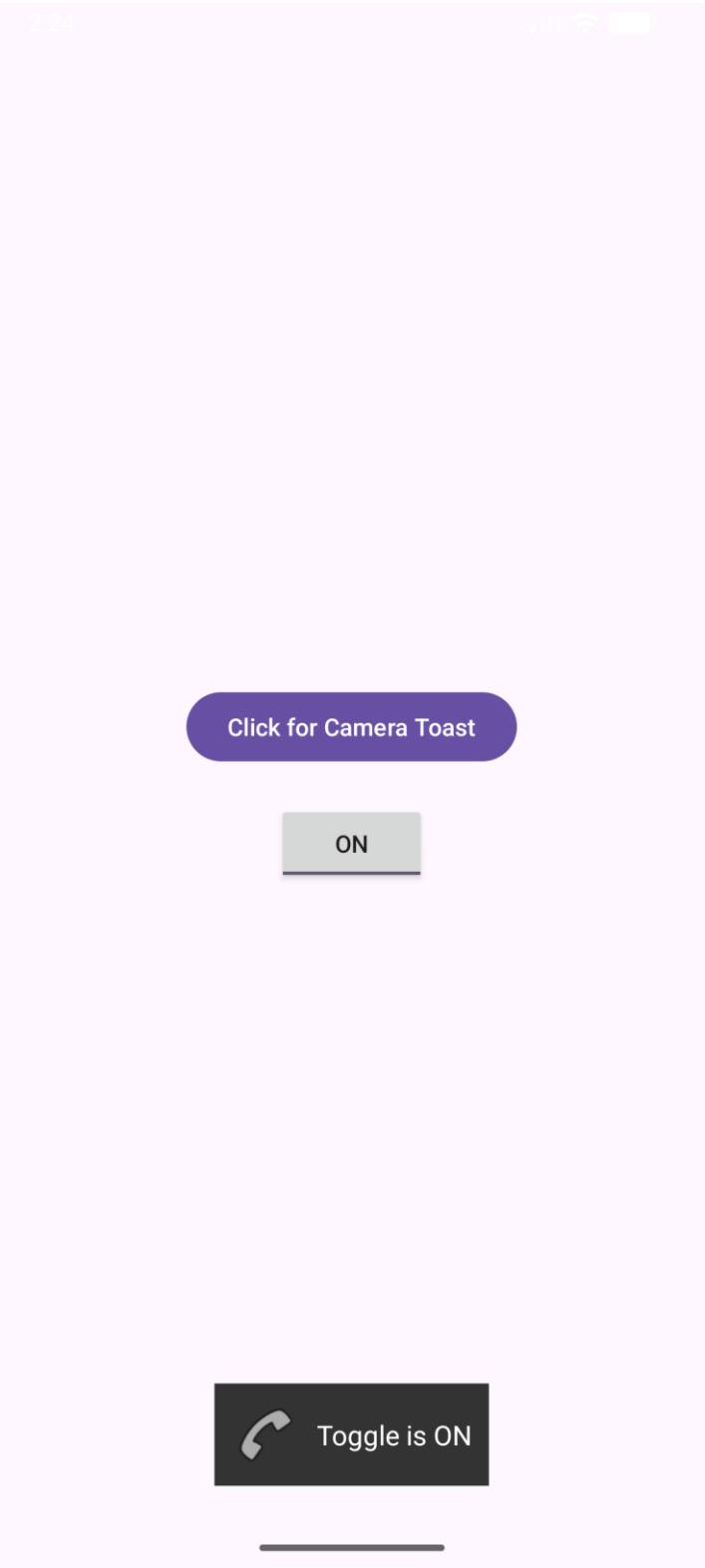
```
package com.example.myapplication;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

public class NewsFragment extends Fragment {
  private String sectionName;

  // Constructor to pass data (Empty constructor is required by Android, so we
overload)
  public NewsFragment() {}

  public NewsFragment(String sectionName) {
    this.sectionName = sectionName;
  }

  @Nullable
  @Override
  public View onCreateView(@NonNull LayoutInflater inflater, @Nullable
ViewGroup container, @Nullable Bundle savedInstanceState) {
    View view = inflater.inflate(R.layout.fragment_news, container, false);

    TextView label = view.findViewById(R.id.sectionLabel);
    if (sectionName != null) {
      label.setText(sectionName);
    }

    return view;
  }
}
```

LAB NO: 4

INPUT CONTROLS IN ANDROID

1. Develop a "Test App" that includes a layout with a Button and a ToggleButton. When each button is clicked, a custom Toast message should be displayed with different images as their content.





activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="16dp">

    <Button
        android:id="@+id/btnStandard"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click for Camera Toast"
        android:layout_marginBottom="20dp"/>

    <ToggleButton
        android:id="@+id/btnToggle"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Toggle Button"
        android:textOn="ON"
        android:textOff="OFF"/>

</LinearLayout>
```

custom_toast.xml:

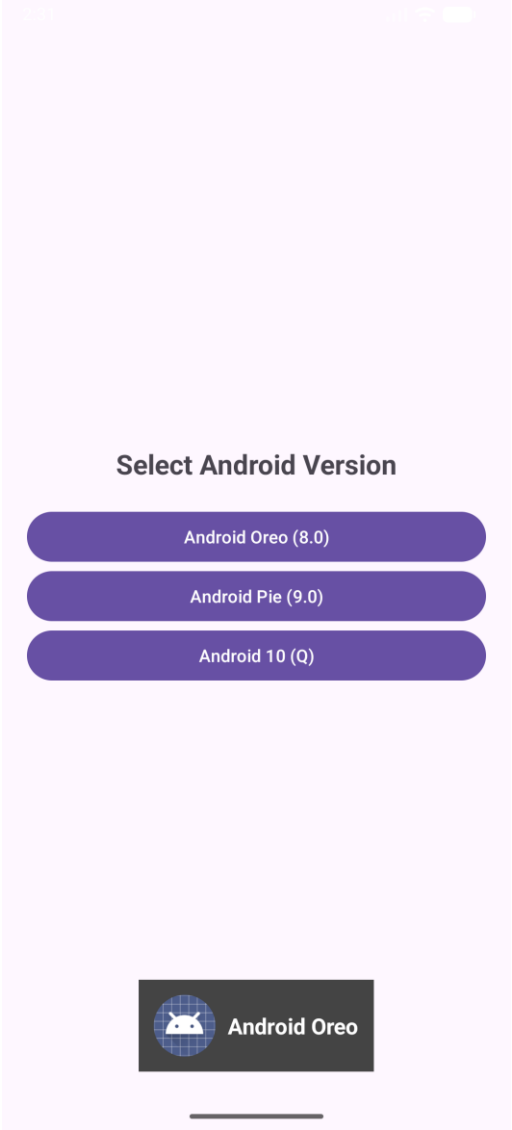
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:id="@+id/custom_toast_container"
  android:layout_width="wrap_content"
  android:layout_height="wrap_content"
  android:orientation="horizontal"
  android:background="#333333"
  android:padding="10dp"
  android:gravity="center_vertical">

  <ImageView
    android:id="@+id/toastImage"
    android:layout_width="40dp"
    android:layout_height="40dp"
    android:layout_marginEnd="10dp"
    android:src="@android:drawable/ic_menu_camera" />

  <TextView
    android:id="@+id/toastText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Custom Toast"
    android:textColor="#FFFFFF"
    android:textSize="16sp"/>

</LinearLayout>
```

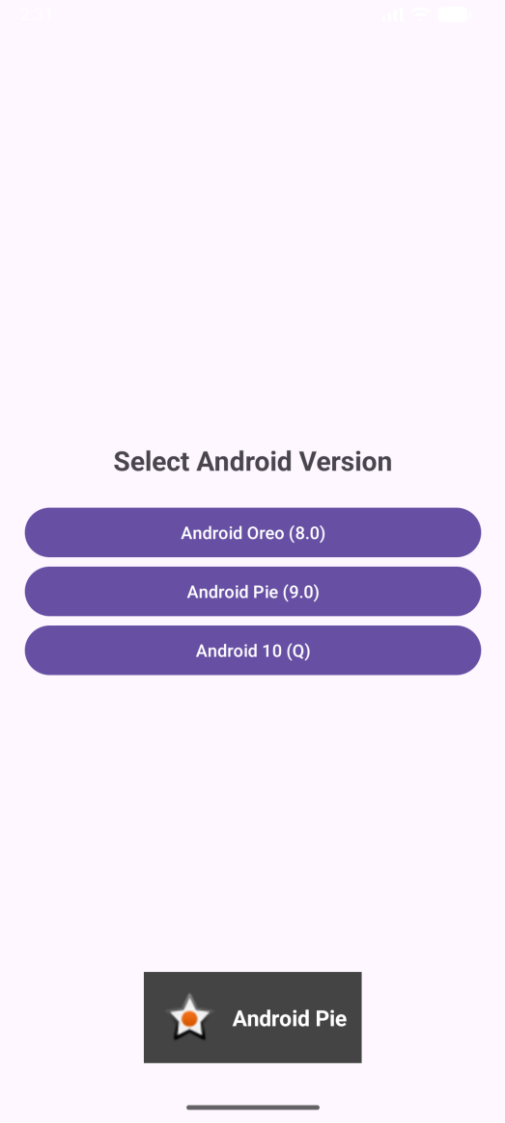
MainActivity.java:

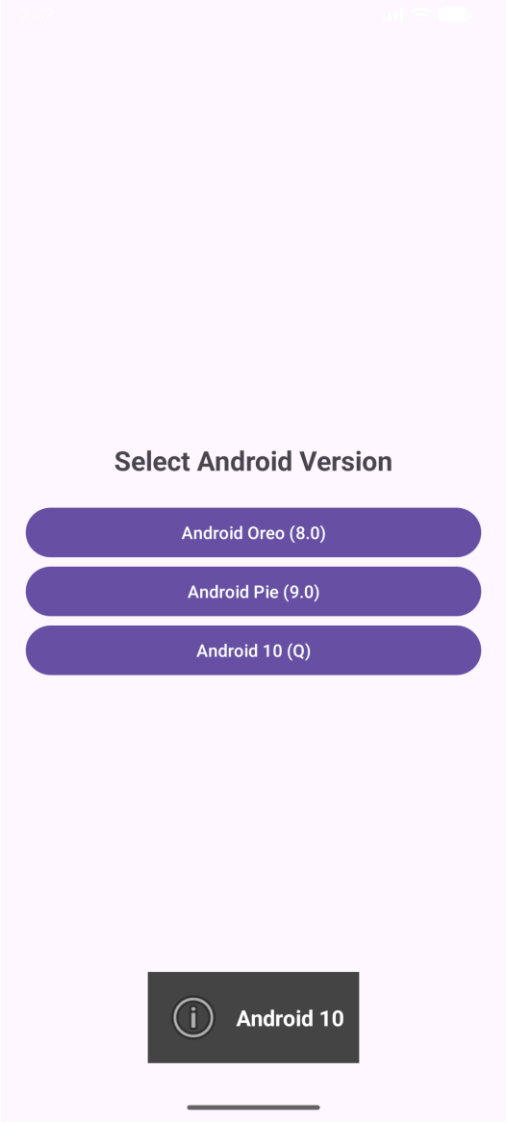
```
package com.example.myapplication;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;
import android.widget.ToggleButton;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    Button btnStandard = findViewById(R.id.btnStandard);
    ToggleButton btnToggle = findViewById(R.id.btnToggle);

    // 1. Standard Button Click Listener
    btnStandard.setOnClickListener(v -> {
      showCustomToast("Camera Action", android.R.drawable.ic_menu_camera);
    });

    // 2. Toggle Button Click Listener
    btnToggle.setOnClickListener(v -> {
      if (btnToggle.isChecked()) {
        showCustomToast("Toggle is ON", android.R.drawable.ic_menu_call);
      } else {
        showCustomToast("Toggle is OFF", android.R.drawable.ic_delete);
      }
    });
  }

  // Helper method to create and show the Custom Toast [cite: 620]
  private void showCustomToast(String message, int imageResId) {
```

```
        // Inflate the custom layout
        LayoutInflater inflater = getLayoutInflater();
        View layout = inflater.inflate(R.layout.custom_toast,
findViewById(R.id.custom_toast_container));

        // Set the text and image
        TextView text = layout.findViewById(R.id.toastText);
        ImageView image = layout.findViewById(R.id.toastImage);

        text.setText(message);
        image.setImageResource(imageResId);

        // Create and show the Toast
        Toast toast = new Toast(getApplicationContext());
        toast.setDuration(Toast.LENGTH_SHORT);
        toast.setView(layout); // Bind the custom layout to the Toast
        toast.show();
    }
}
```

2. Create an app that contains a view with multiple buttons, each labeled with different Android versions. When a button is clicked, a Toast message should appear, displaying the corresponding Android version's name along with its associated icon.







activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
```

```
    android:padding="20dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Android Version"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"/>

    <Button
        android:id="@+id/btnOreo"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Android Oreo (8.0)" />

    <Button
        android:id="@+id/btnPie"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Android Pie (9.0)" />
    <Button
        android:id="@+id/btn10"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Android 10 (Q)" />

</LinearLayout>
```

custom_toast.xml:

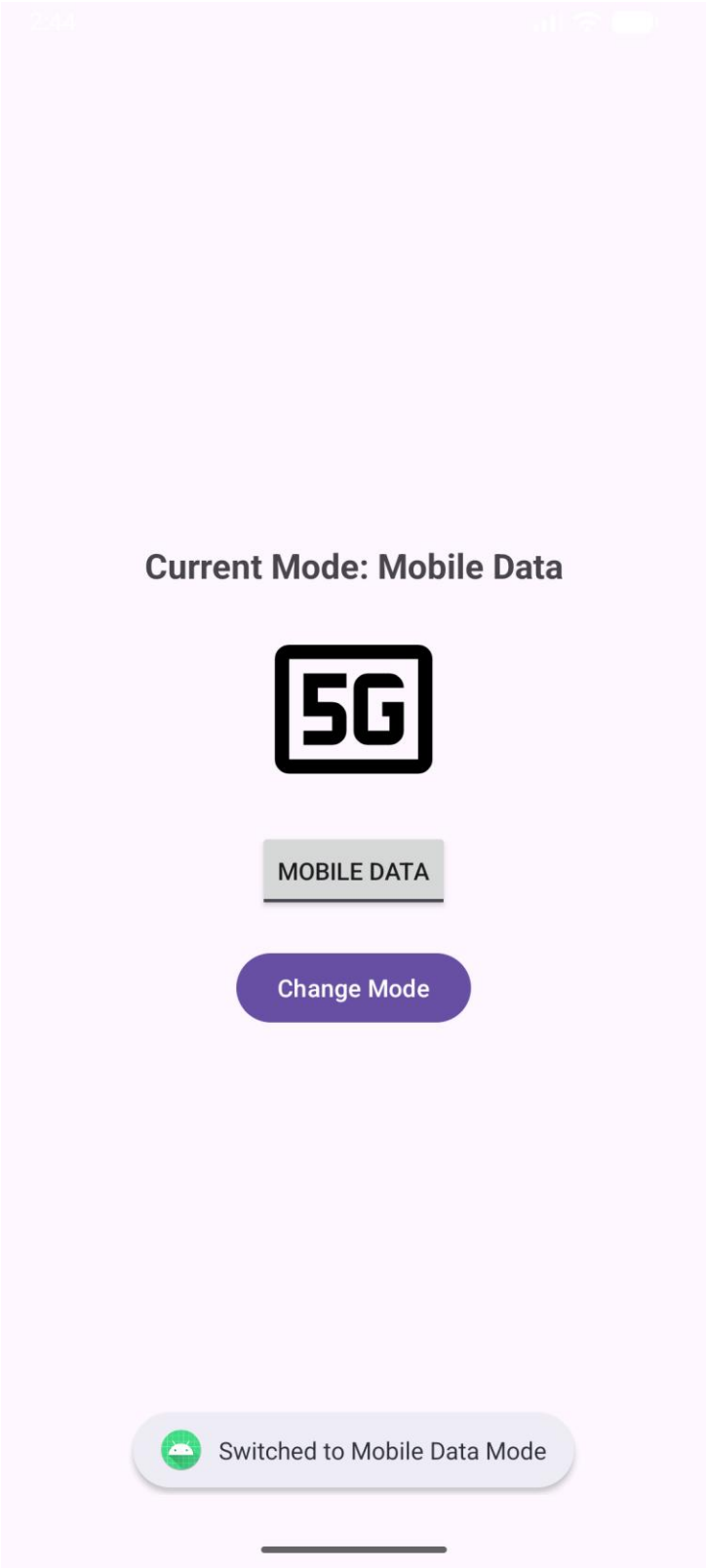
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/custom_toast_container"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:background="#444444"
    android:padding="12dp"
    android:gravity="center_vertical">

    <ImageView
        android:id="@+id/toastIcon"
        android:layout_width="50dp"
        android:layout_height="50dp"
        android:layout_marginEnd="10dp"
        android:contentDescription="Version Icon" />

    <TextView
        android:id="@+id/toastText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textColor="#FFFFFF"
        android:textSize="18sp"
        android:textStyle="bold" />

</LinearLayout>
```

MainActivity.java:

```
package com.example.myapplication;

import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.widget.Button;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    Button btnOreo = findViewById(R.id.btnOreo);
    Button btnPie = findViewById(R.id.btnPie);
    Button btn10 = findViewById(R.id.btn10);

    // Set click listeners for each button
    // Note: Replace 'android.R.drawable.sym_def_app_icon' with your own image
resources
    // e.g., R.drawable.oreo_icon

    btnOreo.setOnClickListener(v ->
        showVersionToast("Android Oreo", android.R.drawable.sym_def_app_icon));

    btnPie.setOnClickListener(v ->
        showVersionToast("Android Pie", android.R.drawable.star_on));

    btn10.setOnClickListener(v ->
        showVersionToast("Android 10", android.R.drawable.ic_menu_info_details));
  }

  // Function to inflate and show the Custom Toast
  private void showVersionToast(String versionName, int iconResId) {
```

```
        // 1. Inflate the Layout
        LayoutInflater inflater = getLayoutInflater();
        View layout = inflater.inflate(R.layout.custom_toast,
findViewById(R.id.custom_toast_container));

        // 2. Set the Text
        TextView text = layout.findViewById(R.id.toastText);
        text.setText(versionName);

        // 3. Set the Image
        ImageView image = layout.findViewById(R.id.toastIcon);
        image.setImageResource(iconResId);

        // 4. Create and Show the Toast
        Toast toast = new Toast(getApplicationContext());
        toast.setDuration(Toast.LENGTH_SHORT);
        toast.setView(layout);
        toast.show();
    }
}
```

3. Develop a view with a ToggleButton labeled "Current Mode" that has two states: "Wi-Fi" and "Mobile Data." Based on the state of the toggle button, an image corresponding to the selected mode should appear, and a Toast message should display the current mode. Additionally, when the user clicks the "Change Mode" button, the app should switch to the corresponding mode and update the image accordingly.



activity_main.xml:

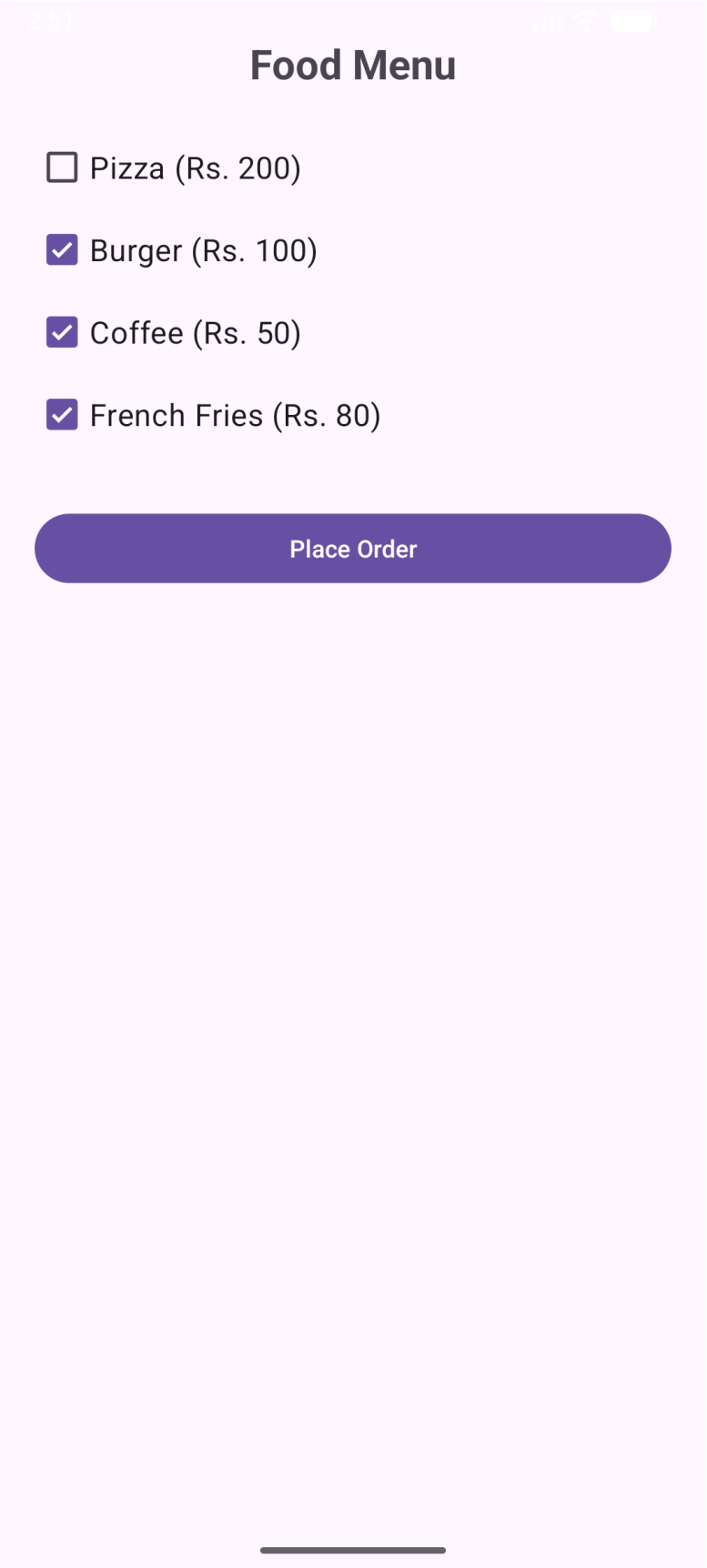
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center"
  android:padding="20dp">

  <TextView
    android:id="@+id/textViewStatus"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Current Mode: Mobile Data"
    android:textSize="20sp"
    android:textStyle="bold"
    android:layout_marginBottom="20dp"/>

  <ImageView
    android:id="@+id/imageViewMode"
    android:layout_width="100dp"
    android:layout_height="100dp"
    android:src="@drawable/ic_mobile_data_foreground"
    android:layout_marginBottom="20dp"/>

  <ToggleButton
    android:id="@+id/toggleButtonMode"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:textOn="Wi-Fi"
    android:textOff="Mobile Data"
    android:checked="false"
    android:layout_marginBottom="20dp"/>
  <Button
    android:id="@+id/buttonChangeMode"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Change Mode" />

</LinearLayout>
```

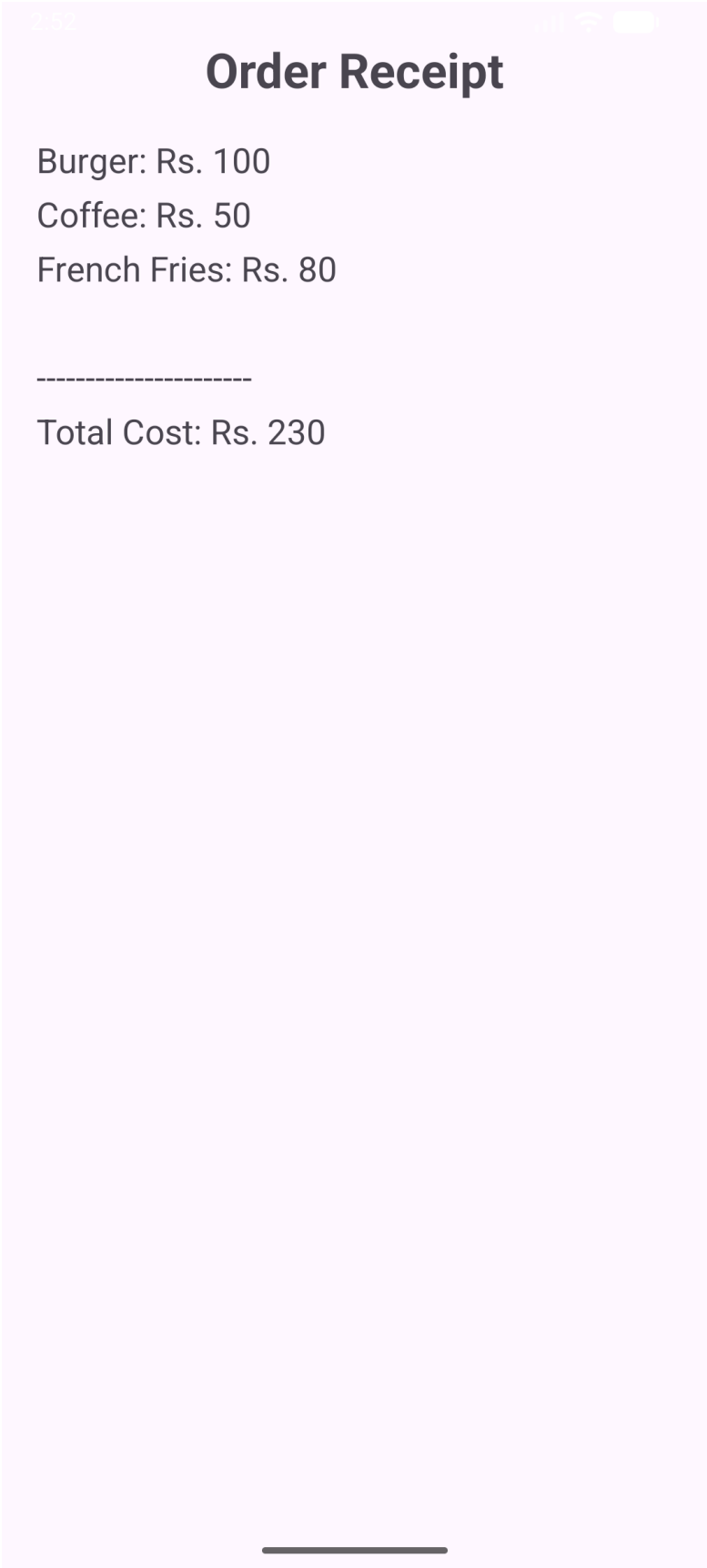
MainActivity.java:

```
package com.example.myapplication;


import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CompoundButton;
import android.widget.ImageView;
import android.widget.TextView;
import android.widget.Toast;
import android.widget.ToggleButton;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  private ToggleButton toggleButtonMode;
  private ImageView imageViewMode;
  private Button buttonChangeMode;
  private TextView textViewStatus;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    // STEP 2: Initialize - Find all views from the layout
    toggleButtonMode = findViewById(R.id.toggleButtonMode);
    imageViewMode = findViewById(R.id.imageViewMode);
    buttonChangeMode = findViewById(R.id.buttonChangeMode);
    textViewStatus = findViewById(R.id.textViewStatus);

    // STEP 3: Set up ToggleButton listener
    toggleButtonMode.setOnCheckedChangeListener(new
CompoundButton.OnCheckedChangeListener() {
      @Override
      public void onCheckedChanged(CompoundButton buttonView, boolean
isChecked) {
        // isChecked = true means Wi-Fi (ON state)
        // isChecked = false means Mobile Data (OFF state)
```

```
            updateMode(isChecked);
        }
    });

    buttonChangeMode.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {
            // Flip the toggle state
            toggleButtonMode.setChecked(!toggleButtonMode.isChecked());
        }
    });

    // Initialize UI based on current state
    updateMode(toggleButtonMode.isChecked());
  }

  private void updateMode(boolean isWifiMode) {
    if (isWifiMode) {
        // Ensure you have ic_wifi_foreground in res/drawable
        imageViewMode.setImageResource(R.drawable.ic_wifi_foreground);
        textViewStatus.setText("Current Mode: Wi-Fi");
        Toast.makeText(this, "Switched to Wi-Fi Mode", Toast.LENGTH_SHORT).show();
    } else {
        // Ensure you have ic_mobile_data_foreground in res/drawable
        imageViewMode.setImageResource(R.drawable.ic_mobile_data_foreground);
        textViewStatus.setText("Current Mode: Mobile Data");
        Toast.makeText(this, "Switched to Mobile Data Mode",
Toast.LENGTH_SHORT).show();
    }
  }
}
```

4. Create a “Food Ordering App” which lists food items with check boxes. Once the user checks /unchecks the item and click on the submit button display the items ordered along with cost of each item and total cost in a new activity. Once the user clicks on the submit button he/she should not be allowed to change the order i.e. he should not be allowed to change the state of the items checked.





activity_main.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="20dp">

  <TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Food Menu"
    android:textSize="24sp"
    android:textStyle="bold"
    android:gravity="center"
    android:layout_marginBottom="20dp"/>

  <CheckBox
    android:id="@+id/cbPizza"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Pizza (Rs. 200)"
    android:textSize="18sp"/>

  <CheckBox
    android:id="@+id/cbBurger"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
```

```
        android:text="Burger (Rs. 100)"
        android:textSize="18sp"/>

    <CheckBox
        android:id="@+id/cbCoffee"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Coffee (Rs. 50)"
        android:textSize="18sp"/>

    <CheckBox
        android:id="@+id/cbFries"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="French Fries (Rs. 80)"
        android:textSize="18sp"/>

    <Button
        android:id="@+id/btnOrder"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Place Order"
        android:layout_marginTop="30dp"/>

</LinearLayout>
```

activity_bill.xml:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp"
    android:gravity="center_horizontal">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Order Receipt"
        android:textSize="28sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"/>

    <TextView
        android:id="@+id/tvBillDetails"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="20sp"
        android:text="Your Bill Details will appear here."
        android:lineSpacingExtra="8dp"/>

</LinearLayout>
```

AndroidManifest.xml (Add the new activity declaration):

```
<activity android:name=".BillActivity" />
```

MainActivity.java:

```
package com.example.myapplication;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    CheckBox cbPizza, cbBurger, cbCoffee, cbFries;
    Button btnOrder;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        // Initialize Views
        cbPizza = findViewById(R.id.cbPizza);
        cbBurger = findViewById(R.id.cbBurger);
        cbCoffee = findViewById(R.id.cbCoffee);
        cbFries = findViewById(R.id.cbFries);
        btnOrder = findViewById(R.id.btnOrder);

        btnOrder.setOnClickListener(v -> {
            StringBuilder bill = new StringBuilder();
            int total = 0;

            // Check items and calculate total
            if (cbPizza.isChecked()) {
                bill.append("Pizza: Rs. 200\n");
                total += 200;
            }
```

```
        if (cbBurger.isChecked()) {
            bill.append("Burger: Rs. 100\n");
            total += 100;
        }
        if (cbCoffee.isChecked()) {
            bill.append("Coffee: Rs. 50\n");
            total += 50;
        }
        if (cbFries.isChecked()) {
            bill.append("French Fries: Rs. 80\n");
            total += 80;
        }

        // Append Total
        bill.append("\n----------------------\n");
        bill.append("Total Cost: Rs. ").append(total);

        // Constraint: Disable controls so user cannot change order
        cbPizza.setEnabled(false);
        cbBurger.setEnabled(false);
        cbCoffee.setEnabled(false);
        cbFries.setEnabled(false);
        btnOrder.setEnabled(false);

        // Navigate to Bill Activity
        if (total > 0) {
            Intent intent = new Intent(MainActivity.this, BillActivity.class);
            intent.putExtra("BILL_DATA", bill.toString());
            startActivity(intent);
        } else {
            Toast.makeText(this, "Please select at least one item",
Toast.LENGTH_SHORT).show();
            // Re-enable if nothing was selected
            cbPizza.setEnabled(true);
            cbBurger.setEnabled(true);
            cbCoffee.setEnabled(true);
            cbFries.setEnabled(true);
            btnOrder.setEnabled(true);
        }
    });
```

```
    }
}
```

BillActivity.java:

```
package com.example.myapplication;


import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class BillActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_bill);

        TextView tvBillDetails = findViewById(R.id.tvBillDetails);

        // Retrieve the bill string passed from MainActivity
        String billData = getIntent().getStringExtra("BILL_DATA");

        if (billData != null) {
            tvBillDetails.setText(billData);
        }
    }
}
```